Real-Time Monitoring and Anomaly Alerting of Power System Meter Driver Data Based on Open Source Hongmeng

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***Abstract*—Conventional meter drive data monitoring and anomaly alarm methods are mainly based on single-point anomaly monitoring, unable to monitor and alarm multiple data at the same time, which affects the stable operation of the power system. Therefore, a real-time meter drive data monitoring and anomaly alarm method based on open source Hongmeng is designed for the power system. To monitor the abnormal load of meter drive data of the power system, the open source Hongmeng system controls the meter drive and monitors the meter data in real time, including voltage, current, power factor, active power, reactive power and other key parameters, to meet the demand for real-time data monitoring. Based on the open source Hongmeng, we generate the alarm function of meter driver data abnormality, trigger the alarm notification service through the system pop-up window, and send key information such as the location of the abnormal meter, the type of abnormality, the value of abnormality, and the time of occurrence to the designated system, so as to ensure the accuracy of abnormality alarms. Comparative experiments are used to verify that the monitoring and alarming accuracy of this method is higher and can be applied in real life.**

Keywords-Open source Hongmeng; Electricity; Meter; Drive data; Real-time monitoring; Anomaly alerting method;

I. INTRODUCTION

The power system is relatively large, including power generation, transmission, substation, distribution, electricity and other aspects, can be converted to primary energy into electricity to meet the demand for electricity. Meter-driven data is the data recorded by meters, such as power meters, electricity meters, fire meters, etc., about the consumption and operation of electric energy. Meter-driven data include metering data, voltage and current data, power and power factor data, time unit data, and abnormal alarm data. All kinds of data in the monitoring of certain difficulties, can not meet the operational needs of the power system. Aiming at this kind of problem, researchers have designed various real-time monitoring and alarm methods. Among them, literature [1] based on the blockchain power system meter drive data real-time monitoring and abnormal alarm method, mainly using the blockchain decentralization, tamper-proof, traceable and other characteristics, to build a safe and reliable meter data monitoring and alarm platform. The distributed storage and encrypted processing of real-time collected power data ensures the integrity and security of the data. Once abnormal data is found, the platform will automatically trigger the alarm mechanism and notify the relevant personnel in time. However, the decentralized nature of this method leads to a relatively slow processing speed, which may not be able to meet the real-time requirements of large-scale power systems.

Literature [2] based on data decomposition and attention mechanism for real-time monitoring and anomaly alerting of meter drive data in power systems decomposes meter drive data according to different time scales or spatial scales, in order to better capture the local features and global trends of the data. However, the method mainly relies on anomaly patterns in historical data for anomaly monitoring. The new anomaly patterns differ greatly from those in the historical data, which may lead to the poor detection effect of the model. Literature [3] based on dual-connection data acquisition system for real-time monitoring of power system meter drive data and anomaly alarm method, using two independent data acquisition channels, respectively connected to the power system meter equipment, to ensure the real-time and reliability of data monitoring. However, the two independent data acquisition channels need to be configured and managed separately, which increases the difficulty of system operation and maintenance. Literature [4] based on the energy distribution data of the power system meter drive data real-time monitoring and anomaly alarm method, through advanced sensors and data acquisition technology, to realize the power system meter drive data real-time collection and monitoring. The collection of energy distribution data may face a certain degree of difficulty, and the data sources and formats of different energy types may differ, which increases the difficulty of data integration. Therefore, this paper combines the advantages of open source Hongmeng and designs a real-time monitoring and anomaly warning method for power system meter drive data.

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II. Design of a Method for Real-Time Monitoring and Anomaly Alerting of Meter-Driven Data in Power Systems Based on the Open Source Hongmeng

A. Monitoring of abnormal loads of meter-driven data in the power system

In the process of real-time monitoring of meter drive data, a large amount of meter drive data is collected for analysis, and the abnormal load characteristics in the meter drive data are extracted so as to judge the operation status of the meter [5-7] . The abnormal load characteristics of meter drive data are shown in Figure 1 below.

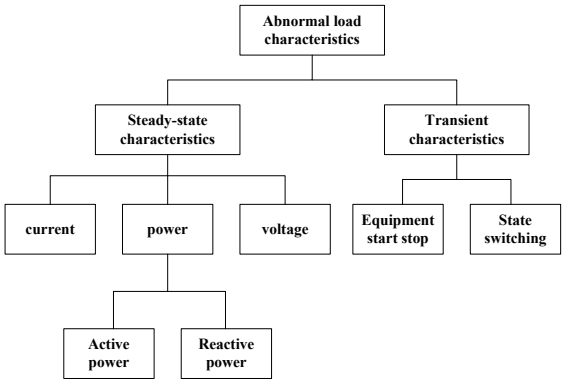

Figure 1. Classification chart of abnormal load characteristics

As shown in Figure 1, the steady state feature can judge the working state of the meter, and the transient feature can judge the load change state. Read the driving data from the meter at a frequency of 15~60min. The power data of power system ammeter is minute level, so it is difficult to capture all abnormal load characteristics of household electrical equipment [8-10] . In this paper, the open source Hongmeng system is connected to the power system, and the operation data of the meter is collected into the system service layer to ensure the comprehensive collection of the meter driving data. After filling and cleaning the collected abnormal load data, standardize and normalize it. The formula is as follows:

$$y = \frac{x-u}{\delta} \tag{1}$$

$$x = \left(x - x_{\min}\right) / \left(x_{\max} - x_{\min}\right) \tag{2}$$

In Eq. (1-2), y is the meter drive data for standardized processing. x is the normalized sample data; u is the expectation of the original distribution of the data; δ is the standard deviation of the driving data; $x_{\min}$ is the minimum data for the current sample; $x_{\max}$ is the maximum data of the current sample [11-13] . The monitoring data were inputted into the open source Hongmeng system to obtain the key parameters with the following equations.

$$\delta_{\max} = \sum_{i=1}^{n} \frac{(IU)_i}{U_i} * \frac{P}{Q} k \tag{3}$$

In equation (3), $\delta_{\max}$ is the key parameters for real-time monitoring of data. U is the voltage. I is the current. k is the power factor; P is the active power; Q is the reactive power [14-16] . Through the open source Hongmeng system to control the meter drive, real-time monitoring of meter data, including voltage, current, power factor, active power, reactive power and other key parameters[17-22], to meet the demand for real-time data monitoring.

$$M(x,y) = \int_{\delta_{\max}} O\left(x(t), y(t)\right) w(t) \tag{4}$$

$$V(x,y) = \int_{\delta_{\max}} \left\| O\left(x(t), y(t)\right) - M(x,y) \right\|^2 w(t) \tag{5}$$

$$F(x,y) = \int_{\delta_{\max}} \left\| O\left(x(t), y(t)\right) \right\|^2 w(t) \tag{6}$$

In Eq. (4-6), $M(x,y)$ is the radial anomaly warning function for meter-driven anomaly data; $x(t)$ is the sample data normalized at time t; $y(t)$ is the abnormal data standardized at time t; $w(t)$ is the weighting function; O is a directional exception parameter. $V(x,y)$ is the outlier anomaly warning function; $F(x,y)$ is the function exception warning function. Normalization process is done for $M(x,y)$, $V(x,y)$, $F(x,y)$, and set its threshold as [0,1], when all three tend to 1, the meter drive data in an abnormal state, display "Alarm!" [23] . When one of the functions tends to 1, and the other functions tend to 0, display "Abnormal!". When the function are close to 0, display "normal". Various types of meter driver data integrated into the open source Hongmeng architecture, as shown in Figure 2 below.

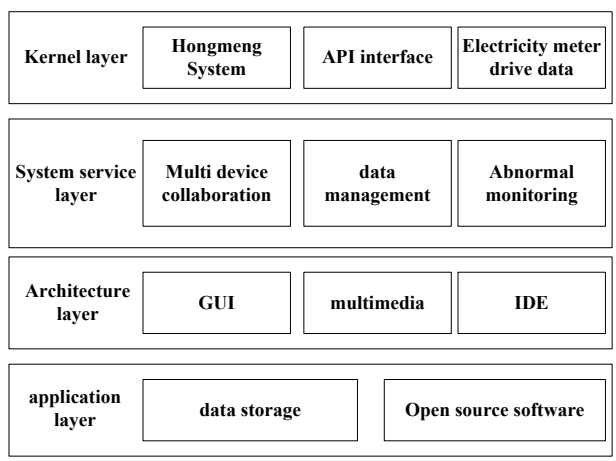

Figure 2. Open source Hongmeng architecture diagram

As shown in Figure 2, open source Hongmeng includes the kernel layer, system service layer, architecture layer, application layer, etc. Among them, the kernel layer can call the power system meter drive data to provide efficient, stable and secure basic services for the open source Hongmeng system. The system can provide various system services through the service layer, including multi device collaboration, data management,

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security assurance, system operation and maintenance, performance optimization and other parts. This layer is the basic condition for Hongmeng cross equipment scheduling and monitoring. With the help of open source software, the performance of Hongmeng system is optimized [24-25] . The architecture layer is the core part of the whole real-time monitoring, which can quickly obtain the real-time monitoring data driven by the meter through the API interface to connect the GUI. The application layer is the top layer of the open source Hongmeng monitoring and alarm architecture, which supports data acquisition, data analysis, data processing and other programs, and can improve the accuracy of real-time monitoring and abnormal alarm of meter driven data. In the system service layer of the open source Hongmeng architecture, obtain the data of the driving power of the electric meter of the power system, as shown in Table 1 below.

TABLE I. POWER SYSTEM METER DRIVE POWER DATA SHEET

Category Number	Station area number	User ID	Electricity meter number	date	Electricity level/kWh
MKL_0	0690100000052	8105360168	0700418018	2022/12/03	644.82
MKL_1	0700100000040	8105360172	0700528018	2022/12/05	816.53
MKL_2	0710100000015	8105360154	0700638018	2022/12/08	883.26
MKL_3	0720100000023	8105360132	0700748018	2022/12/10	859.21
MKL_4	0730100000076	8105360111	0700658018	2022/12/15	745.63
MKL_5	0740100000058	8105360127	0700728018	2022/12/18	571.28
MKL_6	0750100000063	8105360113	0700538018	2022/12/20	436.82
MKL_7	0760100000069	8105360165	0700488018	2022/12/22	428.32
MKL_8	0770100000044	8105360142	0700328018	2022/12/24	299.45
MKL_9	0780100000057	8105360137	0700178018	2022/12/26	578.49

As shown in Table 1, when abnormal data appears in the meter-driven power data table, the alarm notification service is triggered through the system pop-up window, and key information such as the location of the abnormal meter, the type of abnormality, the value of abnormality, and the time of occurrence are sent to the designated system to ensure the comprehensiveness and accuracy of the abnormality alarm.

III. EXPERIMENT

In order to verify whether the method designed in this paper meets the demand of real-time monitoring and anomaly alerting of meter-driven data in electric power systems, this paper conducts experiments to analyze the above methods. The final experimental results are presented in the form of comparison between the monitoring and alerting method based on blockchain in literature [1], the monitoring and alerting method based on data decomposition and attention mechanism in literature [2], and the monitoring and alerting method based on open source Hongmeng designed in this paper. The specific experimental preparation process and the final monitoring and alerting results are shown below.

A. Experimental preparation

This experiment uses the open source Hongmeng system to build an electricity meter data monitoring and alarm experimental environment, to monitor, collect, analyze and process the electricity meter data in the system in real time, and find out the abnormal data in time. Hardware equipment includes Hongmeng system development board, intelligent ammeter supporting Hongmeng system drive, ZigBee and LoRa data transmission module. The software environment includes Hongmeng system development tool chain, data acquisition and analysis software, and alarm notification service. Install Hongmeng system on the development board, start the data acquisition and analysis software, collect the voltage, current and other meter driving data, take the collected data as the actual data, and then obtain the monitoring data. The smaller the error between the two, the higher the accuracy of real-time monitoring. The collected power consumption is shown in Figure 3 below.

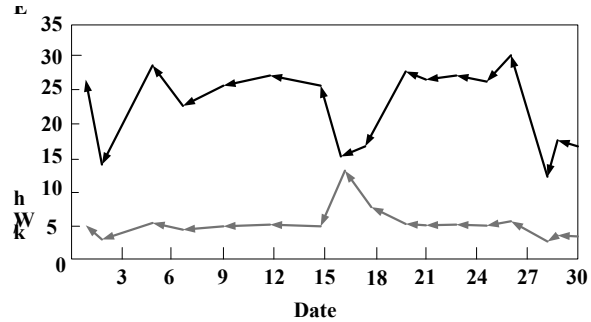

Figure 3. Line graph of electricity consumption

As shown in Figure 3, the black line is the summer electricity consumption; the gray line is the winter electricity consumption. In the open source software, real-time display of meter data changes, and set up real-time monitoring, alarm window, through the data table, curve and other ways to show the meter drive data monitoring results to meet the needs of this experiment.

B. Experimental results

Under the above experimental conditions, this paper randomly selects four types of meter driving data, namely MKL_0, MKL_4, MKL_6, MKL_9, monitor the current and voltage data of each ammeter and give abnormal alarm. When other conditions are known, use the real-time monitoring and abnormal alarm method of electric meter driving data of power system based on open source Hongmeng designed in this paper. The monitoring and alarm conditions are shown in Table 2 below.

TABLE II. TABLE OF CURRENT AND VOLTAGE DATA MONITORING RESULTS

Category Number	phase sequence	Actual current value/A	Monitoring current value/A	Actual voltage value/V	Monitoring voltage value/V	Is there an alarm
MKL_0	A-phase	0.286	0.285	229.521	229.520	Normal display
	B-phase	156.275	156.275	528.965	528.965	Alarm!
	C-phase	0.028	0.028	229.436	229.437	Abnormal!
MKL_4	A-phase	0.384	0.385	128.628	128.627	Abnormal!
	B-phase	0.006	0.006	116.471	116.471	Alarm!
	C-phase	0.028	0.028	227.813	227.812	Abnormal!
MKL_6	A-phase	0.392	0.391	222.454	222.454	Normal display

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	B-phase	0.009	0.009	108.432	108.432	Alarm!
	C-phase	0.345	0.344	225.361	225.362	Normal display
MKL_9	A-phase	112.465	112.465	862.475	862.475	Alarm!
	B-phase	0.001	0.001	101.245	101.245	Alarm!
	C-phase	0.497	0.496	226.471	226.472	Normal display

As shown in Table 2, the error between the actual current and the current monitoring value is within ± 0.1A, and the error between the actual voltage and the voltage monitoring value is within ± 0.1V, which can ensure the accuracy of real-time data monitoring. When the abnormal increase or decrease of current and voltage is detected, the current monitoring condition of the driving data of the meter is prompted by words such as "normal", "abnormal!", "alarm!", etc. It can be seen from the table that the error between the actual current and the current monitoring value is only ± 0.001 A, and the error between the actual voltage and the voltage monitoring value is only ± 0.001 V after using the real-time monitoring and abnormal alarm method of the power system meter driving data designed in this paper based on open source Hongmeng. At the same time, in terms of abnormal current and voltage monitoring, the monitoring values are completely consistent with the actual values. When the current and voltage increase or decrease abnormally, it will be displayed as "alarm!"; When one of the current or voltage increases or decreases abnormally, it is displayed as "abnormal!"; When the current and voltage are normal, it is displayed as "normal". It can be seen that the method designed in this paper can effectively monitor and alarm the normal and abnormal conditions of the meter driving data, and the monitoring and alarm effect is better. When other conditions are consistent, the monitoring and alarm performance based on blockchain in literature [1], the monitoring and alarm performance based on data decomposition and attention mechanism in literature [2], and the monitoring and alarm performance based on open source Hongmeng designed in this paper are compared. The results are shown in Table 3 below.

TABLE III. TABLE OF MONITORING AND ALERTING PERFORMANCE RESULTS

Category Number	Reference [1] is based on a blockchain based monitoring and alarm method			Reference [2] Monitoring and Alarm Method Based on Data Decomposition and Attention Mechanism			The monitoring and alarm method based on open-source HarmonyOS designed in this article		
	Z-score	F1-score	Agj/%	Z-score	F1-score	Agj/%	Z-score	F1-score	Agj/%
MKL_0	0.852	0.898	85.263	0.897	0.925	89.361	0.962	0.999	99.365
MKL_1	0.876	0.902	78.214	0.902	0.936	85.275	0.987	1.000	98.729
MKL_2	0.762	0.815	82.364	0.954	0.982	88.728	0.954	0.972	96.544
MKL_3	0.754	0.802	85.435	0.912	0.947	87.456	0.963	0.982	95.366
MKL_4	0.712	0.782	82.126	0.885	0.901	85.414	0.999	1.000	96.741
MKL_5	0.823	0.864	86.728	0.872	0.894	86.782	0.982	1.000	97.353
MKL_6	0.854	0.892	79.364	0.863	0.885	88.411	0.988	1.000	95.383
MKL_7	0.866	0.903	84.212	0.899	0.921	89.653	0.976	0.999	98.271
MKL_8	0.843	0.886	85.336	0.905	0.954	90.120	0.958	0.985	99.999
MKL_9	0.825	0.867	87.211	0.914	0.947	92.245	0.969	0.976	100.000

As shown in Table 3, Agj is the alarm accuracy rate. Z-score is a standardized value. Set its value range to [0,1]. The closer the value is to 1, the better the standardization is, and the higher the monitoring accuracy is. F1 score is a comprehensive indicator, which can judge the accuracy of real-time monitoring of meter driving data. Its value range is set as [0,1]. The closer the value is to 1, the higher the monitoring accuracy is. When other conditions are consistent, after using the real-time monitoring and abnormal alarm method of electric system meter driving data based on blockchain in literature [1], Z-score is within 0.90, F1 score is within 0.95, and Agj is within 90%. The monitoring and alarm conditions are not ideal, affecting the safe operation of the meter. After using the real-time monitoring and abnormal alarm method of power system meter driving data based on data decomposition and attention mechanism in reference [2], Z-score is within 0.96, F1 score is within 0.99, and Agj is within 95%. The monitoring and alarm performance has been improved, but the fluctuation of each indicator is large, and it is urgent to further optimize it. However, after using the real-time monitoring and abnormal alarm method of electric power system meter driving data designed in this paper based on open source Hongmeng, Z-score is more than 0.95, F1 score is more than 0.97, and Agj is more than 95%, so the accuracy of monitoring and alarm is higher, which meets the research purpose of this paper.

IV. CONCLUSION

In recent years, the smart grid has become an important component of the power system, through informationization, intelligentization, automation and other ways, the meter as the terminal node of the power grid, to ensure the stable operation of the power system. In the process of meter operation, through real-time monitoring of meter drive data, it can analyze the abnormal situation of the power system, so as to meet the operational requirements of the smart grid. Therefore, this paper utilizes the open source Hongmeng to design a real-time monitoring and abnormal alarm method for meter drive data of power system. From the abnormal load monitoring, abnormal alarm function two aspects, automatically obtain the meter drive monitoring data, meter anomaly data. Through the open source Hongmeng cross-platform capabilities, the real-time data monitoring program and the degree of abnormality alarm connected together, timely discovery of anomalies and issue an alarm, the real sense of the meter to achieve intelligent management.

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